

Abraham Elfenbaum

Gameplay/Systems Programmer

West Hartford, CT | 860-836-3590 | a.elfenbaum@gmail.com
[linkedin.com/in/abraham-elfenbaum](https://www.linkedin.com/in/abraham-elfenbaum) | [abrahamelfenbaum.github.io](https://github.com/abrahamelfenbaum)

SUMMARY

Gameplay-focused Software Engineer with 3+ years of professional experience developing core gameplay and engine systems for shipped and in-development games. Strong background in C# and C++ with expertise in gameplay systems, save/load architecture, input handling, UI systems, and performance-conscious logic design. Proven ability to work across C++ and C# within existing codebases on both Unity and Unreal Engine 5, contribute across gameplay, systems, and tools development, and collaborate with designers and artists to maintain high-quality development and QA workflows using industry-standard tools.

TECHNICAL SKILLS

Programming Languages: C#, C++

Development Tools: Unity, Unreal Engine 5, Blueprints, Visual Studio, Git, Perforce

Core Concepts: Object-Oriented Programming (OOP), Application Debugging, Software Engineering Principles, Data Structures and Algorithms, System Architecture, QA Management, Linear Algebra & Mathematics

PROFESSIONAL EXPERIENCE

Software Developer | Digital Scorpion Interactive (Remote) May 2023 - Present

- **Systems Implementation:** Designed and implemented mission-critical gameplay and engine systems, including save/load functionality, persistent data handling, scene and state management, and player control configuration.
- **Cross-Engine Development:** Developed production-ready gameplay features using C# and C++ across Unity and Unreal Engine 5, demonstrating adaptability across multiple engines and frameworks.
- **QA Leadership:** Served as QA Lead on select projects, managing bug tracking pipelines, triaging issues, and coordinating regression testing to ensure gameplay stability and performance.
- **Cross-Discipline Collaboration:** Collaborated closely with designers and artists to integrate and refine complex gameplay features, ensuring alignment between technical systems and creative intent.
- **Team Development:** Mentored junior developers and contributed to coding standard documentation and QA process improvements to support team-wide development practices.

Intern Game Developer | YouROK Corp (Remote) Dec. 2025 – Mar. 2026

- **Feature Development:** Designed and implemented new core gameplay and engine-level features within an existing game project.

- **System Architecture:** Architected a modular game save system supporting local persistence and future migration to cloud-based saving with minimal refactoring.

Gameplay Developer | Improvised Entertainment LLC (Remote) Jan. 2023 – Feb. 2024

- **Gameplay Development:** Designed and implemented real-time gameplay logic and mechanics using C#, with a focus on performance, responsiveness, and maintainability.
- **Code Quality:** Applied Object-Oriented Programming principles to build scalable gameplay systems, improving long-term code health and reducing iteration overhead.
- **Team Workflow:** Contributed to an iterative, team-driven development process, improving development workflows and feature delivery timelines.

RELEVANT PROJECT PORTFOLIO

Project Iron Table | Unreal Engine 5, C++

Designing and building a TTRPG simulator featuring a physics-based dice system with game rules (advantage/disadvantage, private rolls), role-based session management (GM/player permissions), a networked chat system with server and client RPCs, and a custom WYSIWYG rich-text notes editor built as a native Slate widget. All gameplay logic written in C++ with Blueprint used for layout and styling only.

Random Name Generator | C#, .NET 10

Designed and implemented a probabilistic name generator using n-gram language modeling. Architected gender-separated training pipelines and a phonetic pronounceability filter. Developed as a personal project.

Frogdoku | Unreal Engine 5, Blueprint, C++

Developed core game systems including save/load functionality and input binding. Contributed a custom in-game bug reporting tool to streamline QA workflows.

Tales of Valoris: Swallow's Defenders | Unity, C#

Engineered persistent save systems, achievement tracking, and endless mode gameplay logic. Managed QA processes to ensure gameplay stability and performance across builds.

Hit N' Bit | Unity, C#

Developed the game save system, core control logic, and scene management architecture. Acted as QA Lead, overseeing bug reporting, triage, and regression testing.

EDUCATION

Bachelor of Science in Game Development (B.S.) | Full Sail University, Winter Park, FL

- **Accolades:** Salutatorian, Course Director Award
- **Relevant Coursework:** Systems Programming, Data Structures and Algorithms, Software Engineering, Computer Graphics, and Engine Development.